

Geography Ch-14 (Must-Do): Climate Classification - The Koppen System

Tier 1: Must-Do | Source: GC Leong + D.R. Khullar (Koppen scheme) + Majid Husain
Grounded in Qdrant upsc_static book DB + live Current Affairs anchor

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

Koppen's Climate Classification: The Five Groups (A-B-C-D-E) and Their Logic

GS-I (Physical Geography - Climatology) Geography

■ NEWS TRIGGER

Foundation concept (Koppen scheme, as given in D.R. Khullar & GC Leong). Wladimir Koppen devised an empirical classification linking climate to vegetation, using monthly temperature and precipitation. It is the single most exam-relevant climate classification.

■ INTRO & ORIGIN

Koppen (1918) built an empirical, quantitative system: he noticed that natural vegetation is the best expression of climate, so he set temperature and rainfall thresholds that coincide with major vegetation groups. Each type gets a capital letter (A-E) refined by lower-case letters (f, m, w, s, h, c, etc.).

Empirical (based on observed temperature/precipitation data), NOT genetic (not based on causes like air masses). Five major groups correspond to five vegetation groups; a sixth, H (Highland), was later added for mountains.

■ KEY DATA

Group	Name	Defining Criterion
A	Tropical rainy (no cool season)	Coldest month > 18 deg C
B	Dry climate	Evaporation exceeds precipitation (aridity, not temperature)
C	Warm temperate (mild winter)	Coldest month between 18 and -3 deg C; warmest > 10 deg C
D	Cold / microthermal (severe winter)	Coldest month < -3 deg C; warmest > 10 deg C
E	Polar (no warm season)	Warmest month < 10 deg C
H	Highland (added later)	Undifferentiated mountain climate

■ STATIC THEORY

- **Capital letters (1st):** A (tropical), B (dry), C (warm temperate), D (cold), E (polar). B and E are defined by dryness/cold; A, C, D are 'moist' and defined by temperature.
- **Second letters:** for moist climates - f (no dry season, rain all year), m (monsoon, short dry season), w (dry winter), s (dry summer). For dry (B) climates - W (desert/arid), S (steppe/semi-arid).
- **Third letters (temperature of dry & cold climates):** h (hot, mean annual temp > 18 deg C), k (cold); a (hot summer, warmest month > 22 deg C), c (cool short summer).
- **Examples of full codes:** Af = tropical rainforest; Am = tropical monsoon; Aw = savanna (dry winter); BW h = hot desert; BSh = hot steppe; Cs = Mediterranean (dry summer); Cw = China type (dry winter); Cf = maritime; Df = Siberian; ET = tundra; EF = ice cap.
- **Why it dominates exams:** Koppen's codes are precise, map-able and directly link climate to biomes - ideal for objective (Prelims) and diagram-based (Mains) questions.

■ MUST-KNOW FACTS

1. Koppen is EMPIRICAL (data-based), not genetic (cause-based) - a very common Prelims distinction.

■ UPSC TRAPS

- ✗ Koppen's classification is based on the causes of climate (air masses, winds).

2. Group B (dry) is defined by aridity (evaporation > precipitation), NOT by temperature - unlike all other groups. 3. 'f' = no dry season; 'w' = dry winter; 's' = dry summer; 'm' = monsoon (short dry spell). 4. 'h' = hot (mean annual > 18 deg C); 'k' = cold - used with B climates (BWh vs BWk). 5. The 6th group 'H' (Highland) was added later for mountainous, undifferentiated climates.	✓ It is empirical - based on observed temperature and precipitation values; the genetic approach (causes) is a different school. ✗ Group B is a cold climate because it comes after A. ✓ B = DRY climate (deserts + steppes), defined by aridity, and can be hot (BWh) or cold (BWk). ✗ 'w' and 's' both mean wet. ✓ 's' = dry summer (Mediterranean Cs); 'w' = dry winter (savanna Aw / China Cw). Lower-case letters mark the DRY season.
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■ MAINS ANGLE

Evaluate Koppen's classification: its strength (empirical, vegetation-linked, map-able) versus its limitation (ignores causes/air-masses, sharp boundaries where transitions are gradual).

■ STUDY LINK

Geography -> Climatology -> Climate Classification (foundation for all climate-type chapters 15-25)

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA Anchor: Shifting Koppen Zones - Expansion of India's Arid (BWh) & Semi-Arid (BSh) Belts

GS-I
(Climatology) +
GS-III (Climate
Change /
Agriculture)
Geography -
Current Affairs

■ NEWS TRIGGER

2025-26 climate studies using high-resolution Koppen-Geiger maps show India's dry 'B' zones expanding: the arid BWh (Thar) belt creeping eastward/southward and the semi-arid BSh steppe spreading into parts of Maharashtra, Karnataka, Telangana and Andhra Pradesh - a signature of warming + erratic rainfall.

■ INTRO & ORIGIN

Climate classification is not static - as temperatures rise and rainfall becomes erratic, the boundaries between Koppen zones physically migrate. India's 'B' (dry) group is expanding, a direct, map-able consequence of climate change with agricultural and water-security stakes.

Updated Koppen-Geiger maps (1991-2020 baseline + projections to 2050) reveal transition zones. Because Group B is defined by aridity (evaporation > precipitation), warming automatically pushes marginal Aw/BSh land into drier categories.

■ KEY DATA

Region	Old zone (1990s)	New trend (2025/26)
West Rajasthan (Thar)	BWh (arid)	BWh expanding E & S - desertification risk
Central India (MP, Maharashtra)	Aw (savanna)	-> BSh (semi-arid steppe) expansion
Karnataka, Telangana, Andhra	Aw	-> BSh spread - drought management needed

■ STATIC THEORY

- **Group B logic recap:** aridity is judged by a rainfall-temperature threshold (evaporation vs precipitation), so a small rise in temperature can flip a region from moist (A/C) to dry (B) even without a big rainfall drop.
- **Koppen-Geiger maps:** modern high-resolution versions (used by IPCC-era studies) track boundary migration decade-by-decade - the visual evidence of climate change.
- **Policy link:** National Action Plan on Climate Change (NAPCC) + state action plans now realign crop planning, drought preparedness and water management to updated climate maps.

■ MUST-KNOW FACTS

1. Group B (dry) expansion in India = warming + erratic rainfall tipping Aw/BSH land into drier categories.
2. Thar's BWh belt is projected to expand eastward and southward; BSh steppe spreading into peninsular interior.
3. Studies suggest India's dry zone could grow 10-15% in area by 2050 on current trends.
4. Koppen 'B' is aridity-defined, so temperature rise alone can shift the zone boundary.
5. Feeds NAPCC / state climate action plans for crop and water re-planning.

■ UPSC TRAPS

- ✗ Koppen zones are permanent lines on a map.
- ✓ They are threshold-based; as temperature/precipitation change, the boundaries physically migrate over decades.
- ✗ Only rainfall decline can expand a desert (BWh).
- ✓ Because B is aridity-defined (evaporation vs rain), rising temperature increases evaporation and can expand the arid zone even with stable rainfall.

■ MAINS ANGLE

Use shifting Koppen boundaries as concrete, map-able evidence of climate change in India, and link the expanding 'B' (dry) belt to desertification, food security and adaptation policy.

■ STUDY LINK

Geography -> Climate Classification (Koppen) + GS-III climate change; connect to desertification (Ch-7 Advanced) and NAPCC